Structural-based engineering of virus-like particles formed by endogenous human gag-like proteins

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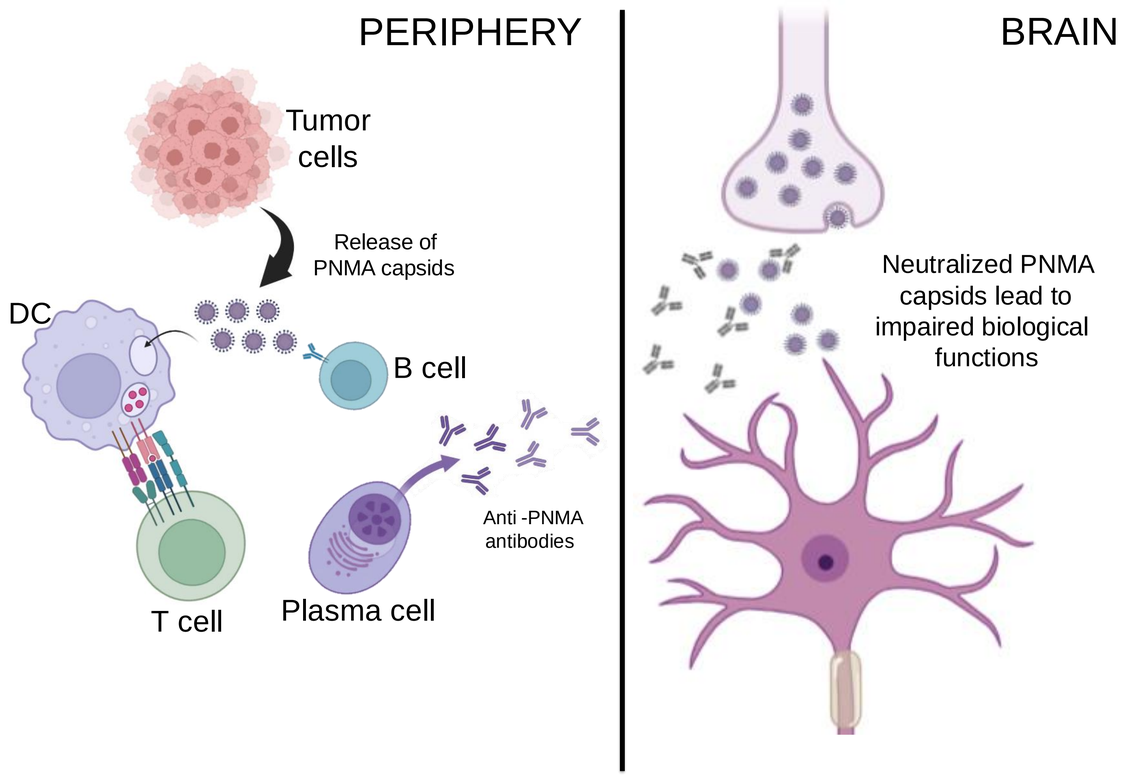
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Background

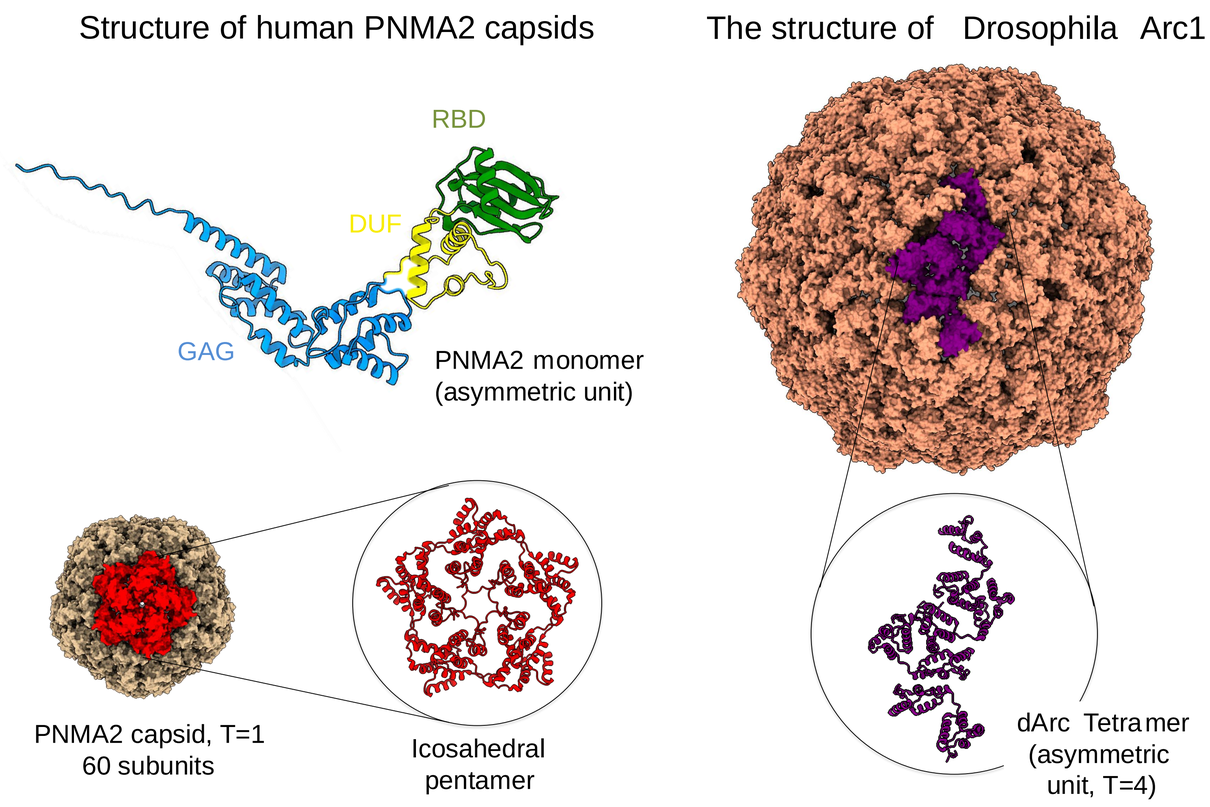
Gag genes, which encode structural proteins, have been domesticated for important host functions, and domesticated gag-like proteins exist in humans. Several of these proteins have been shown to form virus-like particles (VLPs) that play crucial roles in cellular communication and neuronal plasticity. However, although many other gag-like proteins also preserve complete capsid domains, their potential VLP structure and function remains unclear. In our group, we aim to use single-particle cryo-EM to obtain the structures of VLPs formed by several domesticated human gag-like proteins and to employ AI-driven protein design tools to engineer these assemblies for biomedical applications.

Human gag-like proteins are involved in cancer and neurological processes



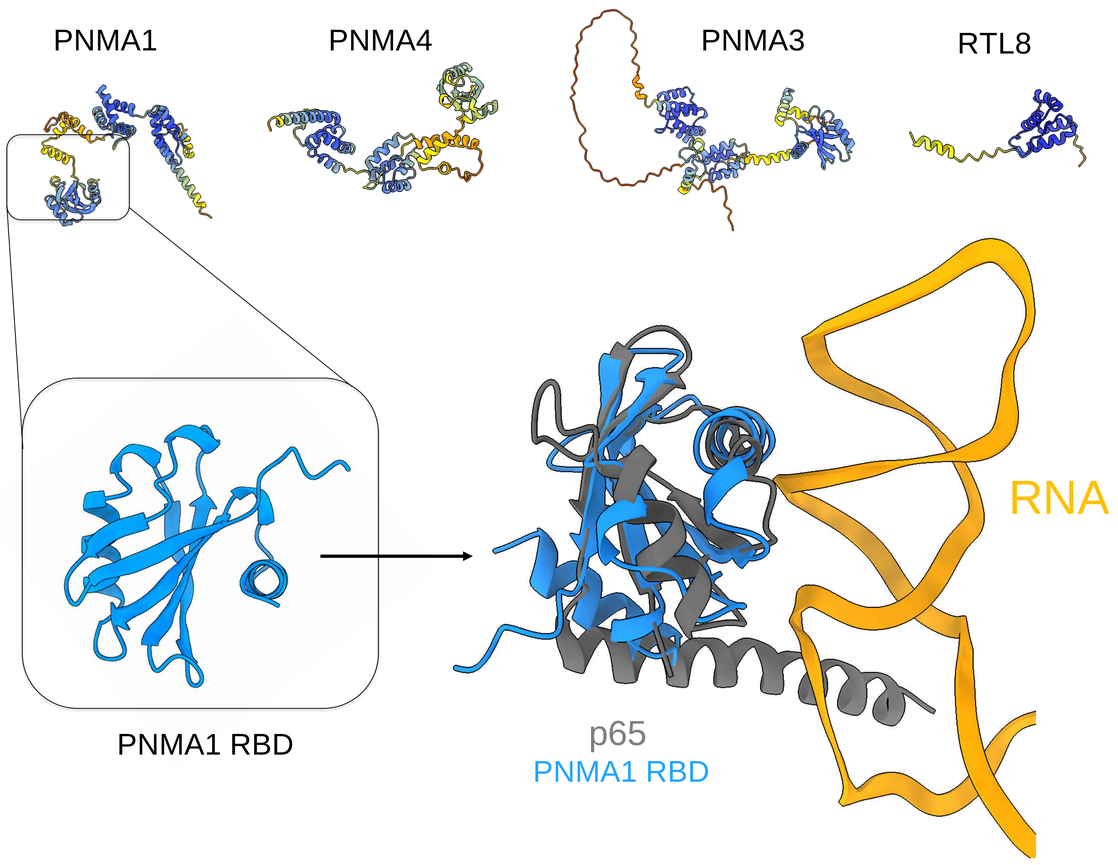
Cancer cells may express gag-like proteins ectopically, releasing capsids that promote an adaptive autoimmune response with the generation of autoantibodies. These autoantibodies can later cross the blood-brain barrier and block the physiological function of these capsids in the brain. DC, dendritic cell.

Cryo-EM can resolve the structure of capsids formed by gag-like proteins



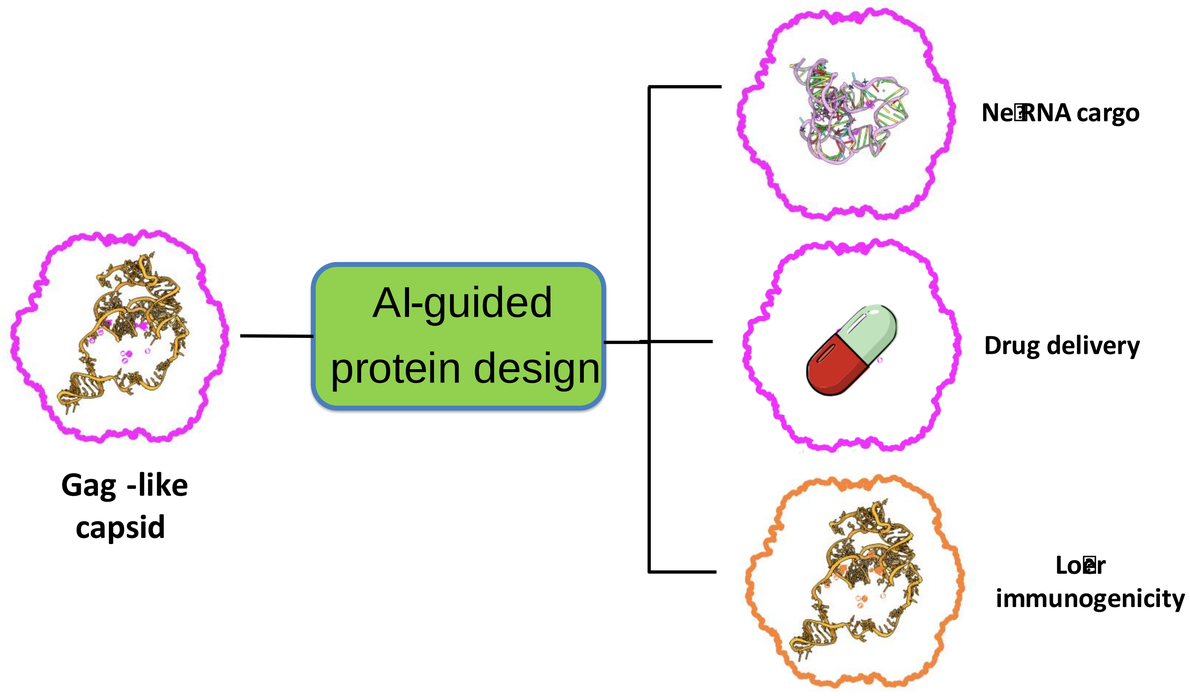
The structures of human PNMA2 (PDB 8RBJ) and Drosophila Arc (dArc, PDB 8PAP) capsids have been solved by cryo-EM. Both type of proteins form icosahedral capsids but with different architectures. While PNMA2 forms T=1 capsids (60 subunits), dArc forms more complex T=4 capsids (240 subunits).

Human gag-like proteins possess a putative RNA-binding domain (RBD)



The AF2-predicted structures of human PNMA and RTL genes reveal the presence of a gag-like domain and several types of N- and C-terminal domains. Remarkably, some PNMA genes like PNMA1, 3 and 4 possess an N-terminal domain which shares structure similarity with well-known RNA-binding domains. In the image, the putative PNMA1 RNA-binding domain, shown in blue, is aligned to the T. thermophila RNA-binding domain of the telomerase La-related protein p65 (grey) bound to telomerase RNA (orange, PDB 7LMA).

AI-powered design of gag-like capsids



The structural information of various gag-like capsids will serve as foundational scaffolds for the AI-guided design of novel protein-based capsids. By leveraging these naturally occurring architectures, we will engineer synthetic capsids with enhanced or entirely new functionalities. Specifically, these redesigned capsids will enable the selective and efficient packaging of custom RNA cargo, facilitate the encapsulation and targeted delivery of therapeutic compounds, and be optimized to reduce immunogenic responses. To achieve these goals, we will employ advanced computational tools such as RFdiffusion, ProteinMPNN, NanoBodyBuilder-AI and AF2.

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